

MSME IDEA HACKATHON 6.0Reference No. :- **26INC06KL047695**

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6. Theme	Industry 4.0 and 5.0
7. Idea Sector	Miscellaneous Sector (Environment, Forests, Water & Sanitation; Foods, Beverages, FMCG, Consumer Goods; Infrastructure, Construction, Housing; IT, ITES, Electronics, White Goods, Telecommunication; Metals, Engineering, Machinery, Automation and Transportation, Automotive, E Vehicles, Railways, Aviation, UAV and any other sub-sector)
8. Title of proposed idea/innovation	AeroSense Autonomous Environmental Monitoring Telemetry UAV Platform
9. Briefly explain newness/uniqueness of the innovation	AeroSense is a multi-layered autonomous UAV platform designed to bridge precision flight control with real-time IoT environmental sensing. Unlike traditional drones that act purely as camera rigs or single-purpose tools, AeroSense implements a dual-processing architecture that physically separates flight dynamics from sensor computation. Key Unique Features: Dual-Processor Architecture: Powered by a SpeedyBee flight controller running ArduPilot alongside an ESP32-S3 companion computer, preventing telemetry and sensor tasks from overloading primary flight algorithms. Real-Time MAVLink Obstacle Avoidance: Integrated ultrasonic rangefinder array connected via the ESP32 gateway that dynamically feeds MAVLink distance vectors to ArduPilot for automated collision braking and position holding. Long-Range IoT Telemetry Logging: Features a LoRa wireless telemetry link for ultra-long-range remote data streaming, paired with onboard MicroSD logging for secure data redundancy in remote environments. Multi-Parametric Sensing Suite: Equipped with a integrated modular payload (HC-SR04, BME280, PMS5003, MQ-135) for real-time air quality, dust density, and environmental atmospheric profiling. Robust Power Infrastructure: Built on a customized F450 framework utilizing dedicated power distribution architecture and dual buck regulators for clean power isolation between heavy motor draw and sensitive payload sensors.
10. Concept & Objective	AeroSense is designed as an autonomous, dual-processor UAV system that merges long-range IoT atmospheric sensing with ArduPilot flight stabilization. By offloading sensor telemetry and MAVLink proximity messaging to a dedicated ESP32-S3 companion computer, the platform maintains deterministic flight safety while executing continuous environmental analysis. Objectives: Develop a reliable dual-processor architecture combining an ArduPilot flight controller (SpeedyBee) with an ESP32 companion unit. Integrate real-time obstacle avoidance utilizing ultrasonic rangefinder inputs processed through a custom MAVLink MAV_DISTANCE_SENSOR gateway. Stream multi-parametric atmospheric metrics (particulate matter, air quality index, temperature, humidity) over long-range wireless links (LoRa) while retaining local MicroSD backup logs. Ensure stable field operation using custom power distribution, dual buck regulation, and an F450 multirotor frame setup.

11. Specify the potential areas of application in industry/market in brief	Industrial Environmental Safety Monitoring: Autonomous monitoring of ambient air quality, toxic gases (MQ-135), and particulate emissions (PMS5003) around chemical plants, construction sites, and manufacturing hubs without risking human personnel. Smart Agriculture Forestry: Profiling atmospheric humidity, micro-climates, and regional ambient data over agricultural plots to assist in crop management and localized micro-climate tracking. Disaster Response Search Support: Deploying long-range LoRa-linked autonomous survey units during hazardous events (fires, gas leaks) to assess environmental conditions and obstacle barriers ahead of rescue teams. Infrastructure Urban Perimeter Inspection: Automated indoor and outdoor structural scanning utilizing ultrasonic MAVLink distance-braking to inspect bridges, high-rise facades, and restricted zones safely.
12. Briefly provide the market potential of idea/innovation	The global commercial UAV and environmental monitoring markets are growing rapidly due to stricter environmental regulations and rising demand for autonomous industrial inspection tools. AeroSense addresses a clear market gap: traditional industrial sensing drones are high-cost and proprietary, whereas low-cost options lack processing capacity and active collision avoidance. Key Commercial Advantages: Cost Efficiency: Built on standard, accessible components (F450, SpeedyBee, ESP32), lowering manufacturing and maintenance costs by up to 60 compared to legacy enterprise platforms. Modular Payload Capabilities: The dual-processor system allows rapid adaptation for specialized sensors (thermal, multispectral, gas specific) without requiring firmware re-engineering on the flight controller. Scalable IoT Integration: Built-in LoRa capabilities align directly with Smart City frameworks and Industry 4.0 IoT telemetry pipelines, enabling multi-drone fleet monitoring across large geographical areas.
13. Uploaded Block diagram/ flow chart/ Circuit Diagram/Pictures	View/Download